2.4 GHz FMCW Radar — Baseband Circuit Simulation

David Brown

October 2026

The baseband circuitry (video amplifier and filtering, half-rail reference, power input, sync and IF input) was simulated in NI Multisim 14.2 using library parts, one block per card and then all of it on one sheet (card 8). Nothing has been built yet. Plots are in Section 3.

1 Results

Card	Check	Result	Requirement	
1 Video amp	Outputs at half rail (DC)	1.07 mV off	≤ 150 mV	**PASS**
	Gain at 159 Hz / 1 kHz / 15.9 kHz	48.7 / 54.5 / 50.9 dB	48.7 / 54.5 / 50.6 dB	**PASS**
	0.5 mV, 1 kHz in	0.266 V out	0.27 V	**PASS**
	First power-up gain, 10 mV in	106.8 mV out	107 mV	**PASS**
	Clipping	± 4.06 V	about ± 4.2 V	**PASS**
	Two-tone: 417 / 870 Hz lines	19.1 / 21.1 mV	19.0 / 21.2 mV	**PASS**
2 Half-rail ref	Divider / buffered output	5.70 / 5.58 V	5.70 / 5.58–5.72 V	**PASS**
	150 kHz ripple reaching vref	-162 dB	≤ -100 dB	**PASS**
3 12 V input	Protected / analog rail	11.72 / 11.62 V	11.68 / 11.58 V	**PASS**
	Ripple on analog rail	3.2 mVpp	about 4 mVpp	**PASS**
4 Reverse supply	-12 V applied	-11 mV gets through	< 0.5 V	**PASS**
5 Sync divider	Sync pulse level	275 mV	270–300 mV	**PASS**
6 IF input	At 417 Hz / 2.44 GHz	-6.6 / -57.7 dB	-6.6 / -57.7 dB	**PASS**
7 Scaled radar	Beat tone at 5 / 10 / 15 m	435.0 / 870.7 / 1305.8 Hz	435.2 / 870.4 / 1305.6 Hz	**PASS**
	Beat tone with leakage on	872.7 Hz	stays at 870 Hz	**PASS**
8 All blocks (as built)	Outputs at half rail (DC)	4.36 V off	≤ 150 mV	**FAIL**
8 All blocks (fixed)	Outputs at half rail (DC)	1.08 mV off	≤ 150 mV	**PASS**
	Gain vs. card 1 (C6 = 47 µF)	6.4 dB low	within 0.3 dB	**FAIL**
	Gain vs. card 1 (C6 = 3.3 mF)	0.15 dB	within 0.3 dB	**PASS**
	Two-tone 870 Hz line	21.0 mV	21.2 mV	**PASS**
	150 kHz ripple at output	0.06 µV	< 1 mV	**PASS**
	Power-up from 6 V	settles in 0.53 s	< 1 s	**PASS**

Changes made on card 8:

- Wired U1A's $V-$ and U1B's $V+$ pins. They were left unconnected when the blocks were copied onto one sheet, which latched the amp at the rail.

- C6 raised from 47 µF to 3.3 mF. The half-rail reference was moving with the signal and the first stage amplified it.

Notes:

- Card 8's analog rail reads 11.49 V because the TI op-amp model draws 5.7 mA per amp (datasheet: 1.4 mA typ.). Expected about 11.6 V on the real circuit.
- The op-amp model can't run below about 4.4 V supply, so power-up starts at 6 V.

2 Summary

All cards pass. Card 8 passes after the wiring fix and the C6 change.

3 Figures

Card 1: video amp

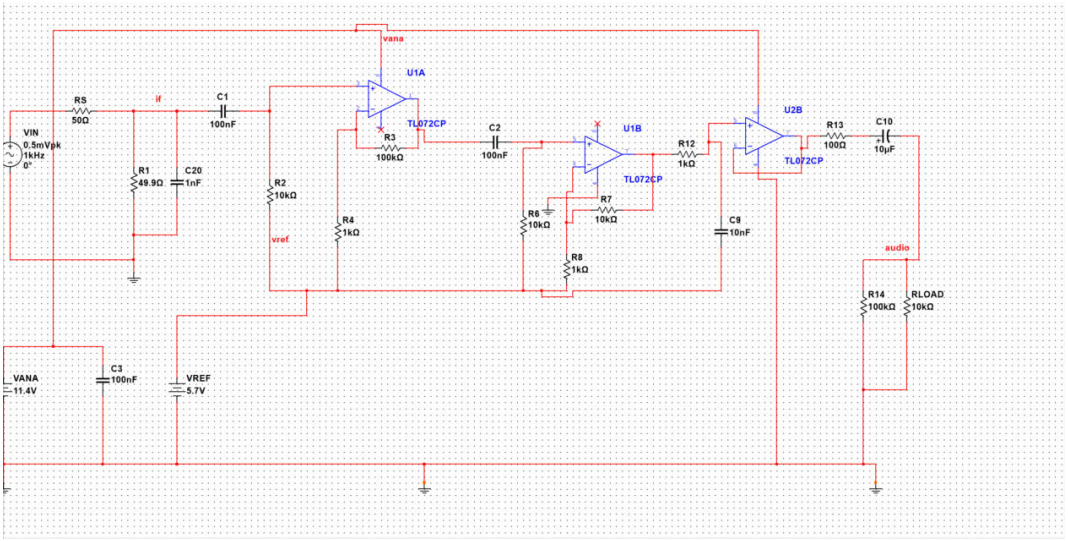


Figure 1: Multisim schematic

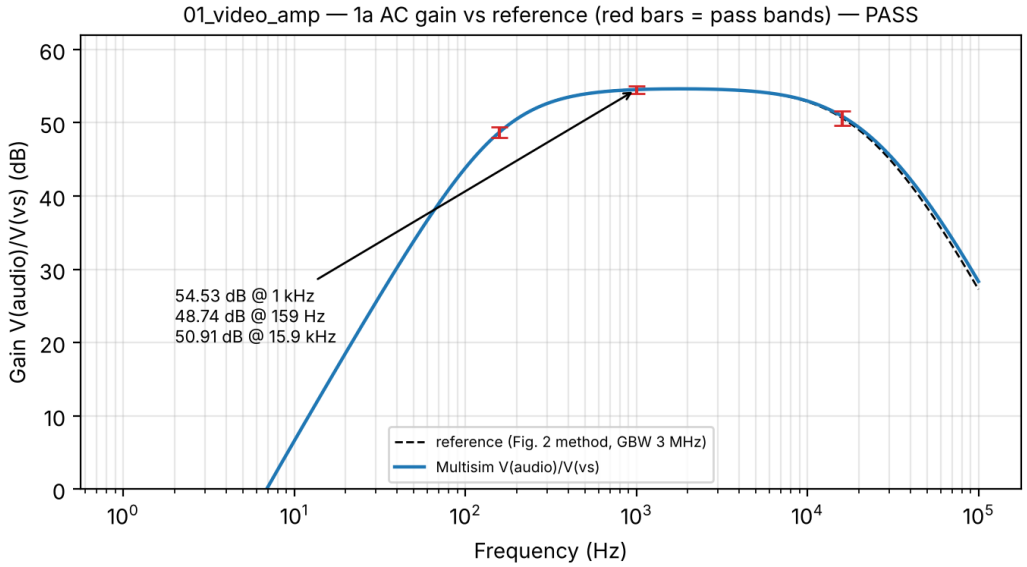


Figure 2: Gain vs. frequency

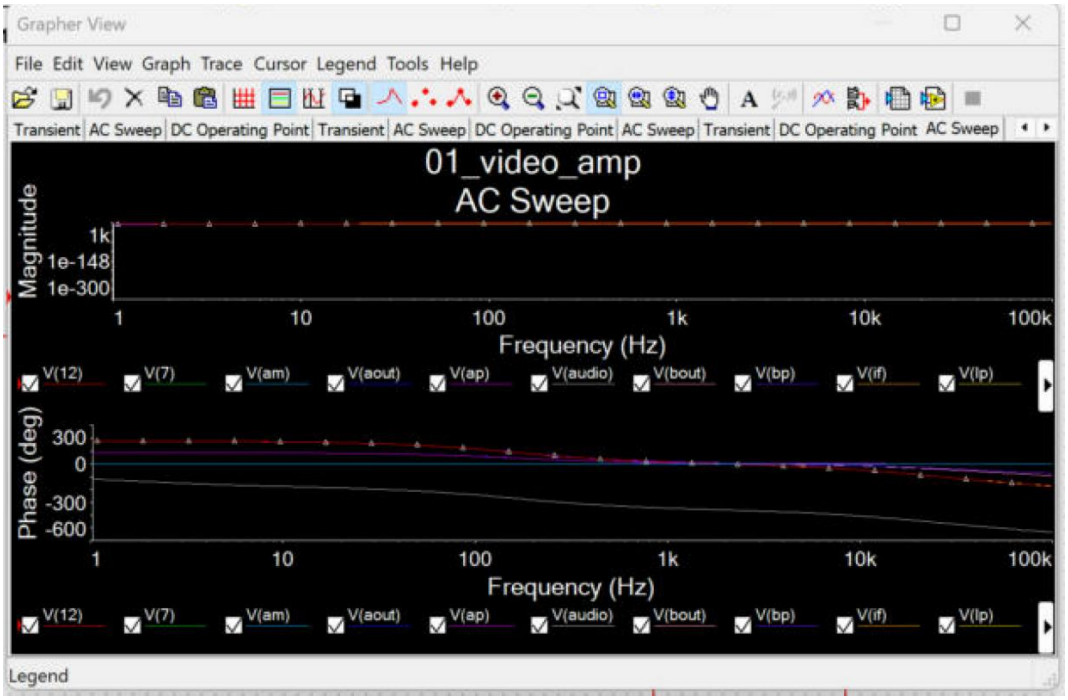


Figure 3: Multisim grapher, AC sweep

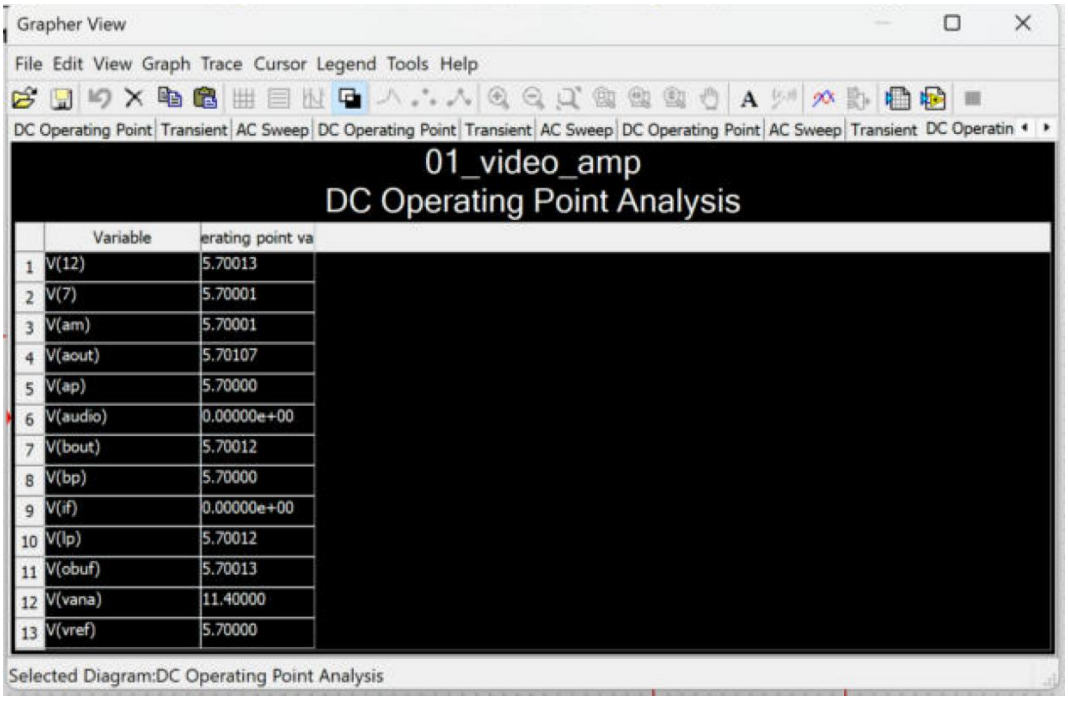


Figure 4: Multisim grapher, DC operating point

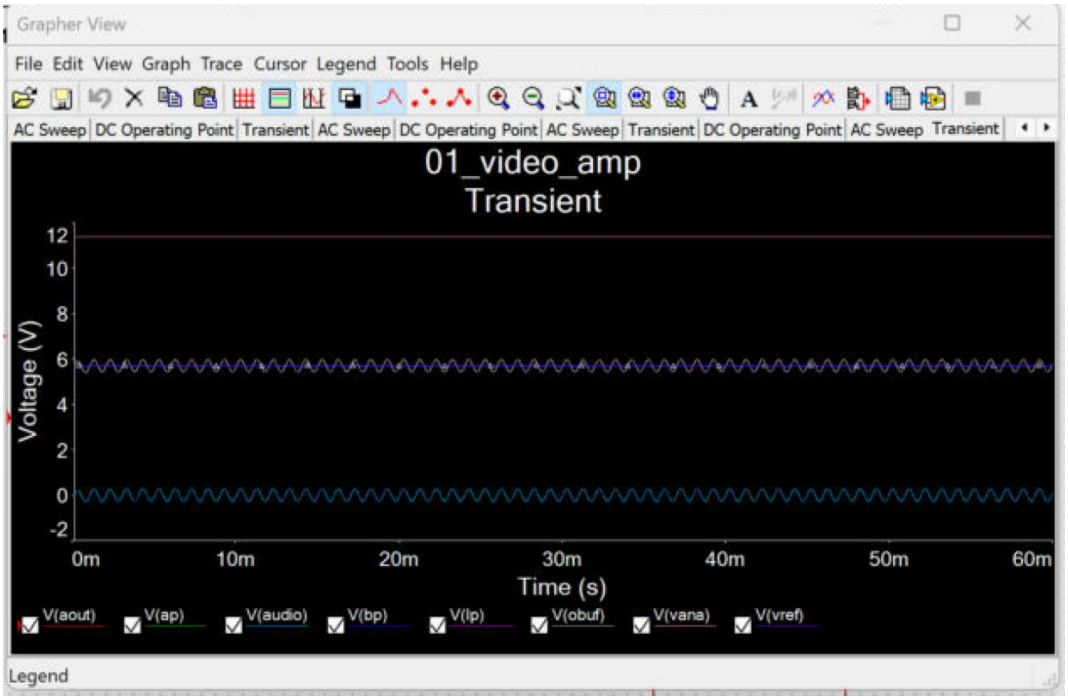


Figure 5: Multisim grapher, transient

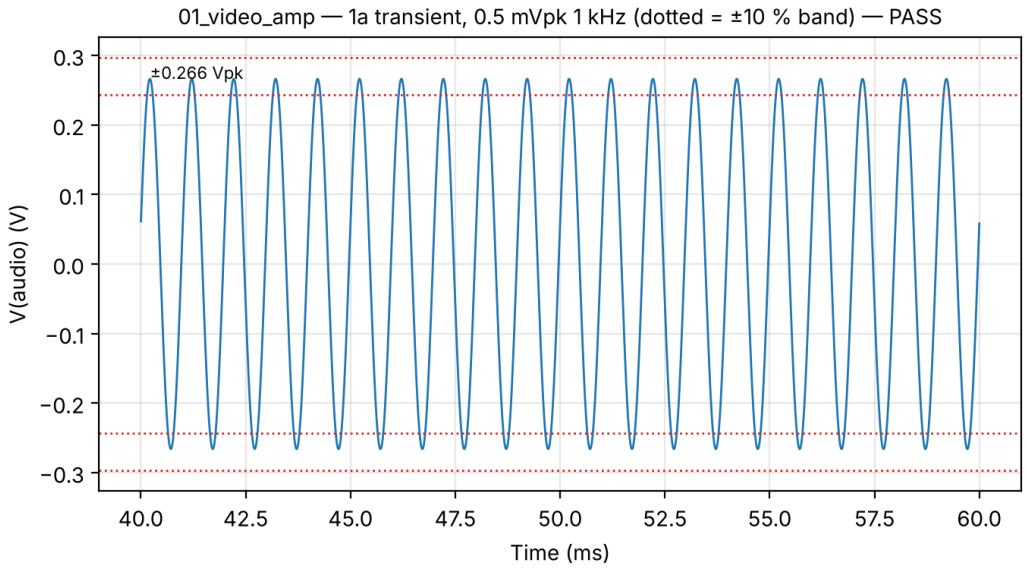


Figure 6: 1 kHz transient

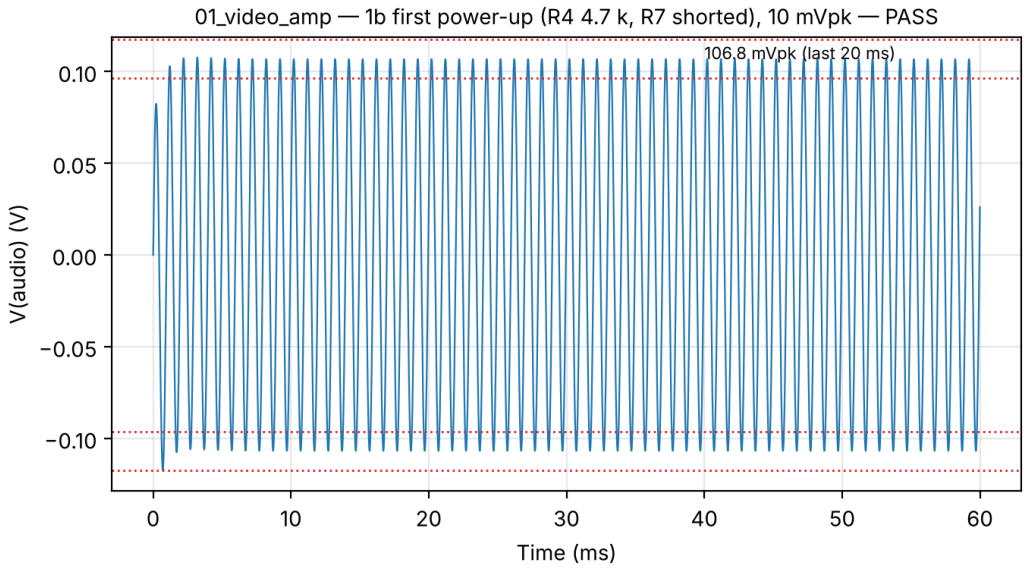


Figure 7: First power-up gain

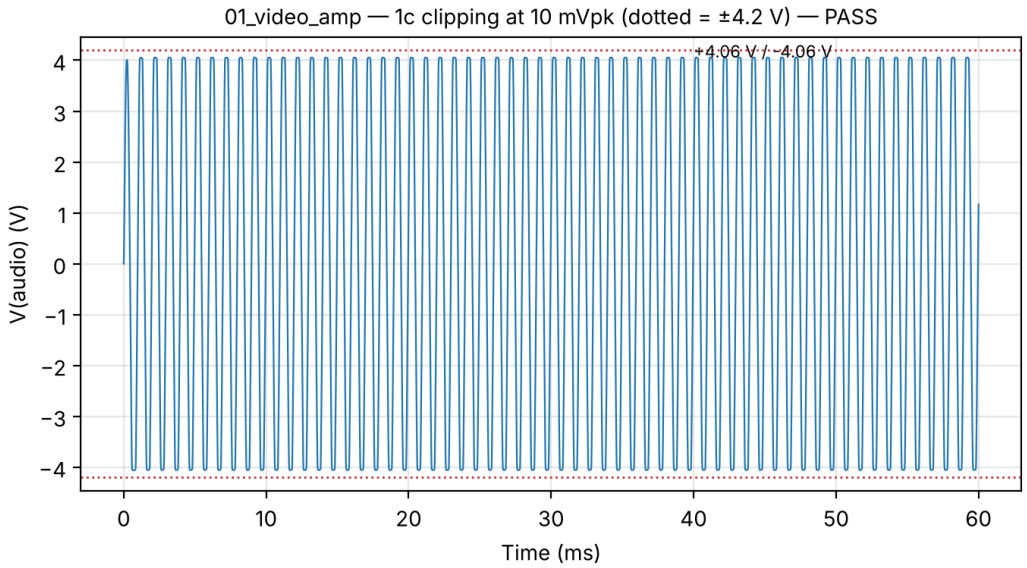


Figure 8: Clipping at 10 mV in

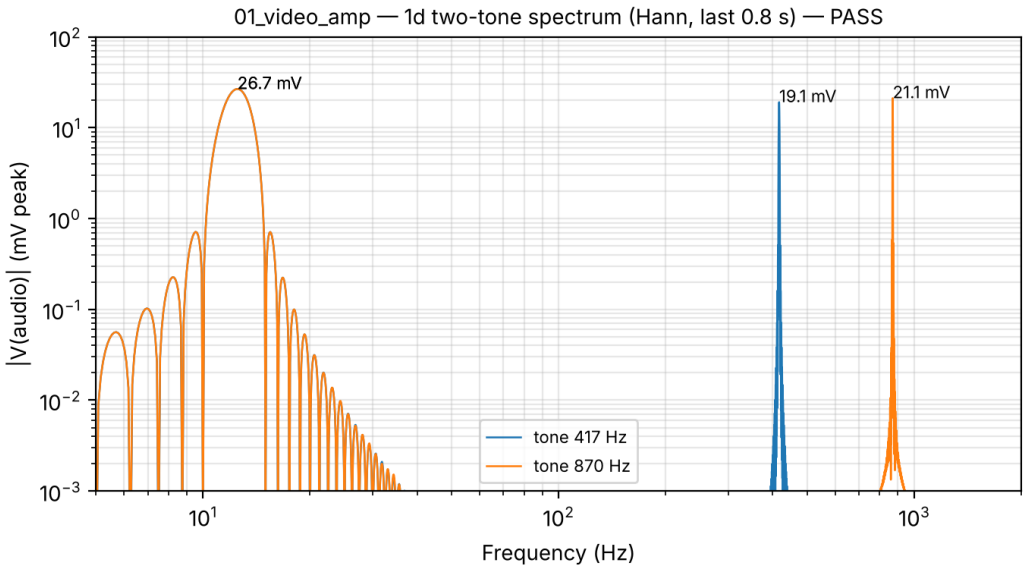


Figure 9: Two-tone spectrum (leakage + echo)

Card 2: half-rail reference

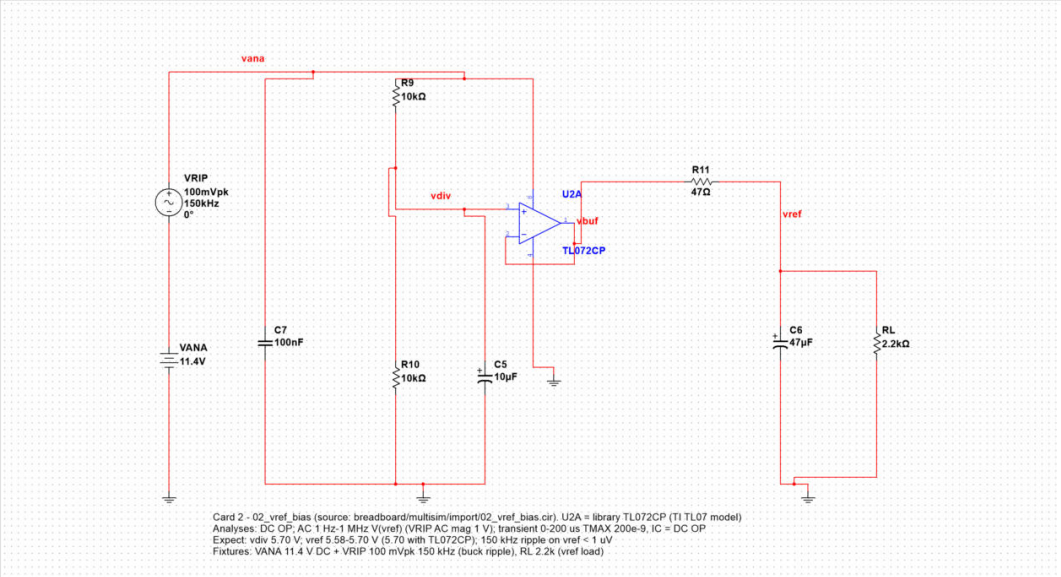


Figure 10: Multisim schematic

02_vref_bias — DC operating point — PASS

node	DC operating point
V(vana)	11.40000 V
V(vbuf)	5.70001 V
V(vdc)	11.40000 V
V(vdiv)	5.70000 V
V(vref)	5.58078 V

Figure 11: DC operating point

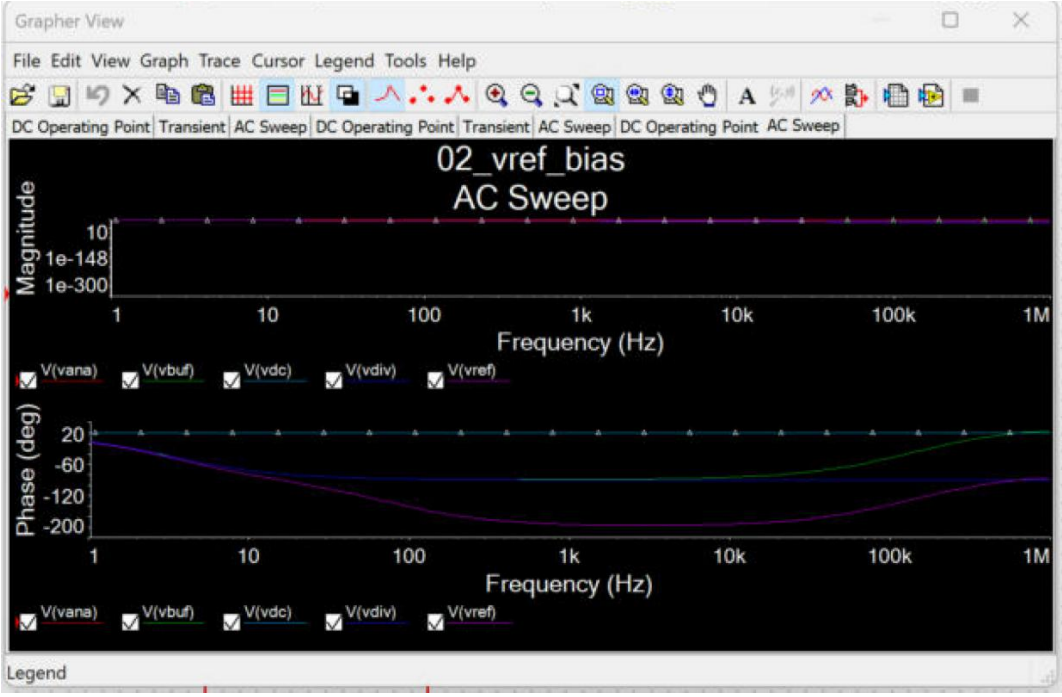


Figure 12: Multisim grapher, AC sweep

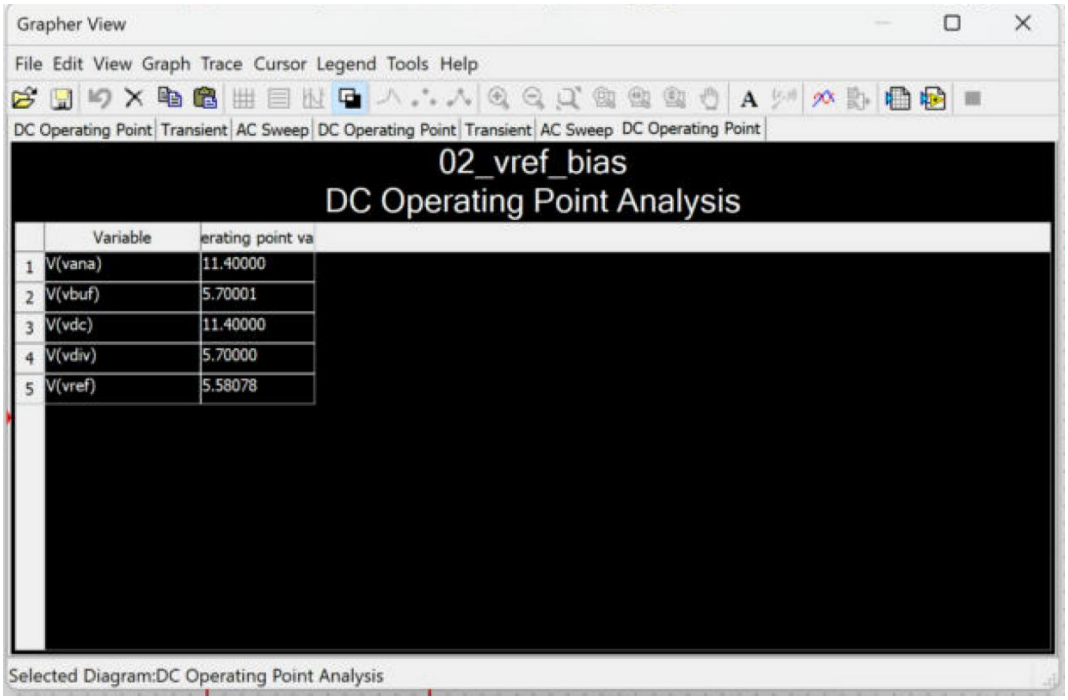


Figure 13: Multisim grapher, DC operating point

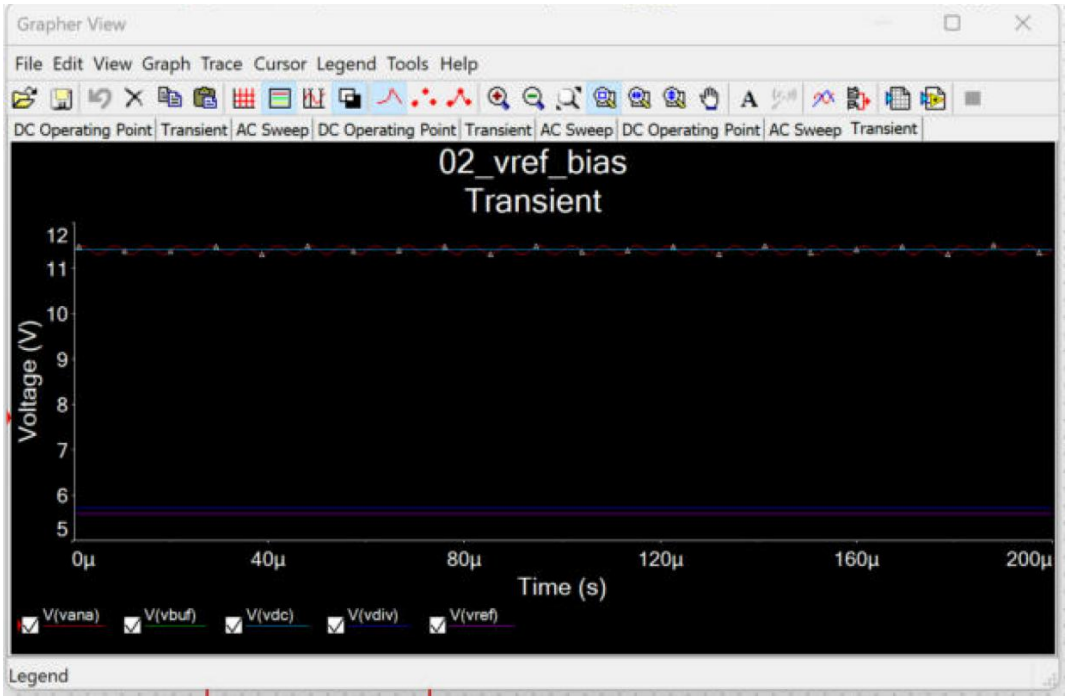


Figure 14: Multisim grapher, transient

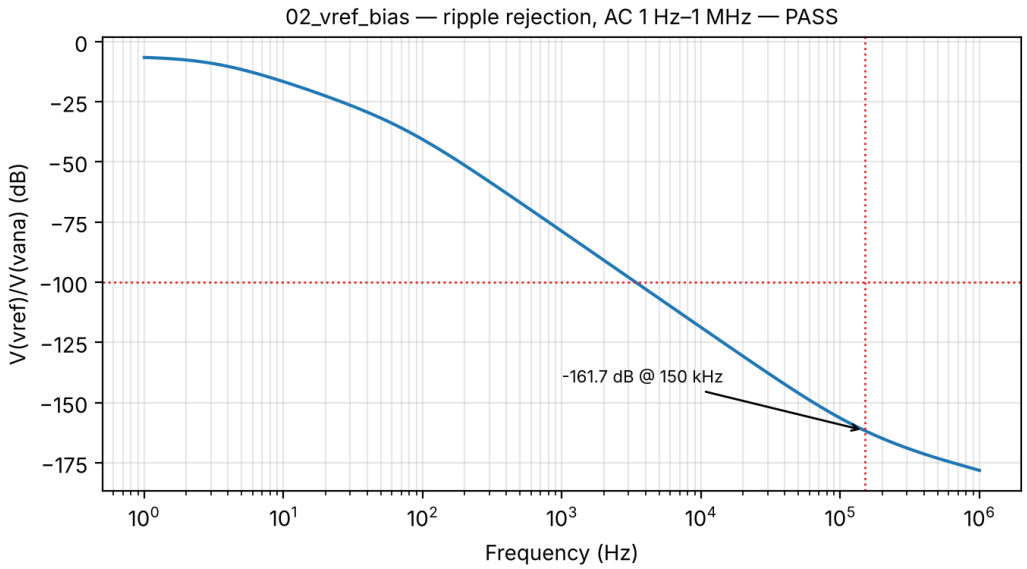


Figure 15: 150 kHz rejection to vref

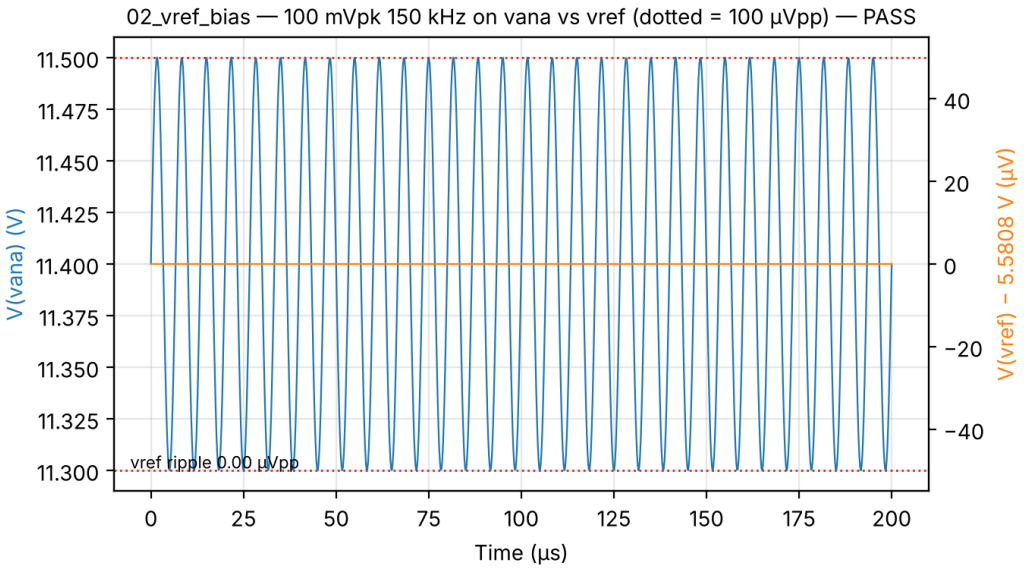


Figure 16: Ripple on vref

Card 3: 12 V input

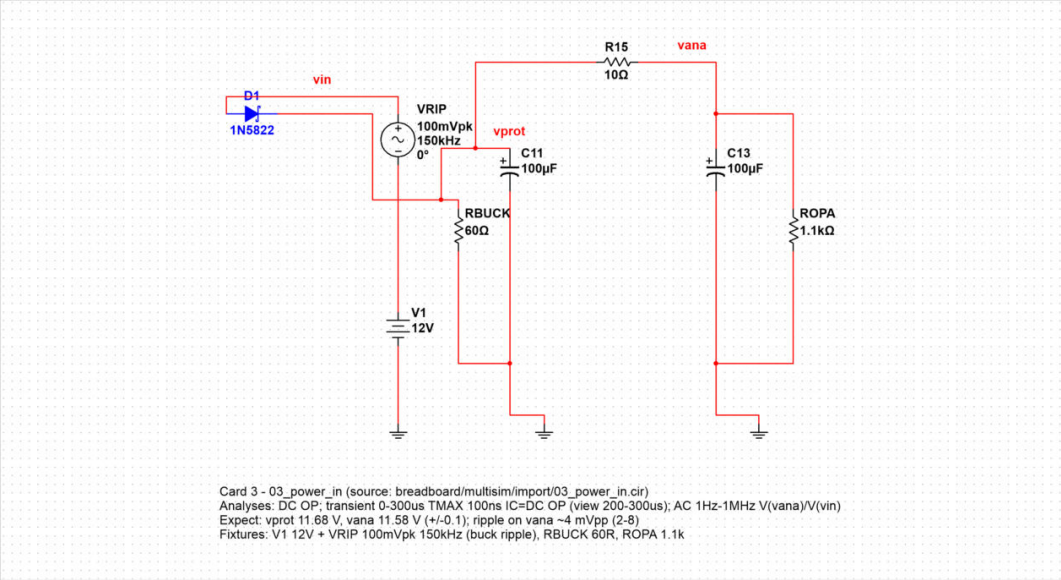


Figure 17: Multisim schematic

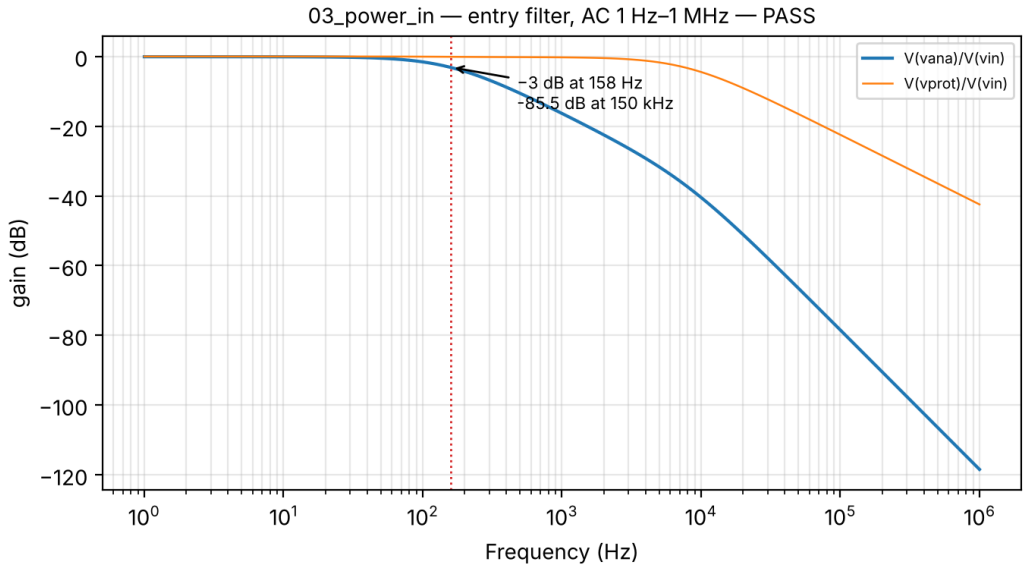


Figure 18: Input filter response

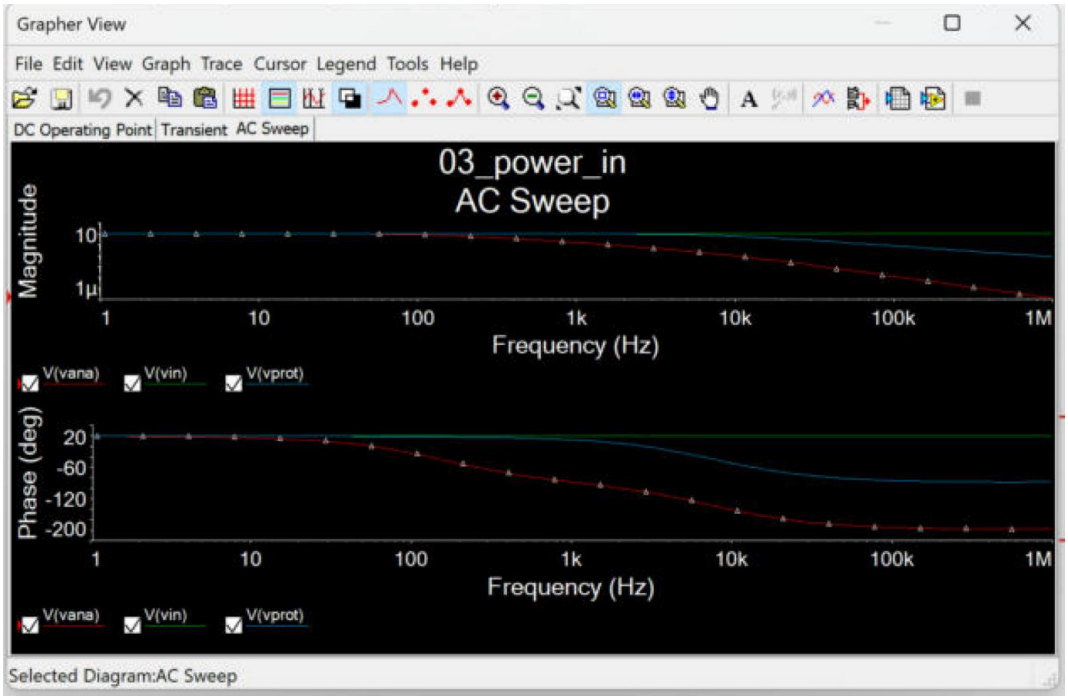


Figure 19: Multisim grapher, AC sweep

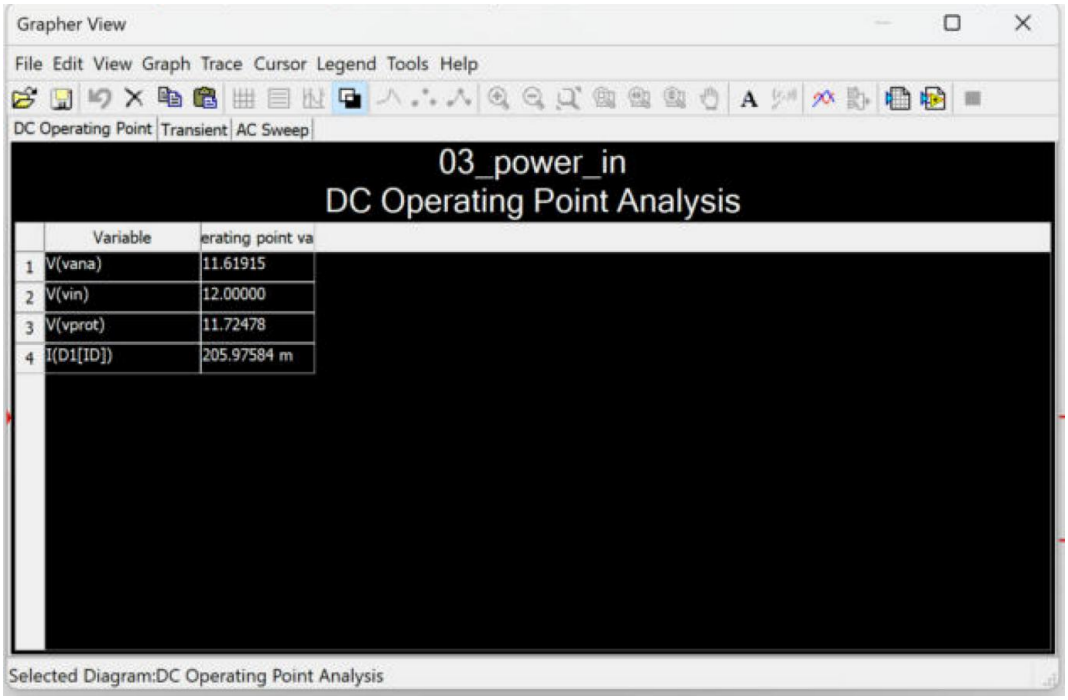


Figure 20: Multisim grapher, DC operating point

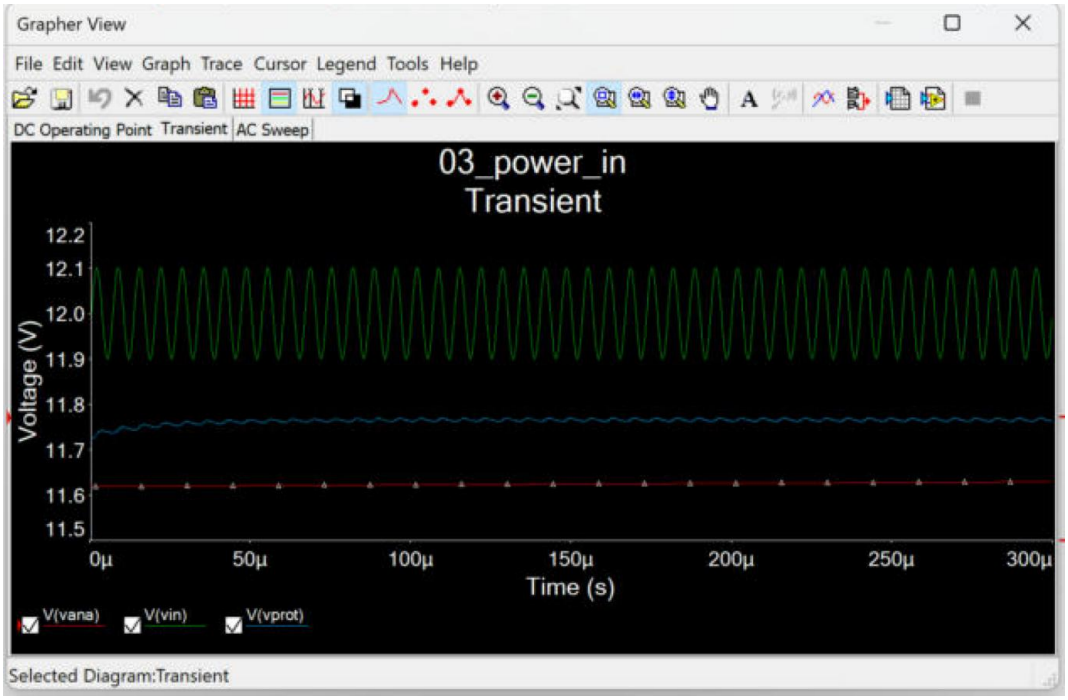


Figure 21: Multisim grapher, transient

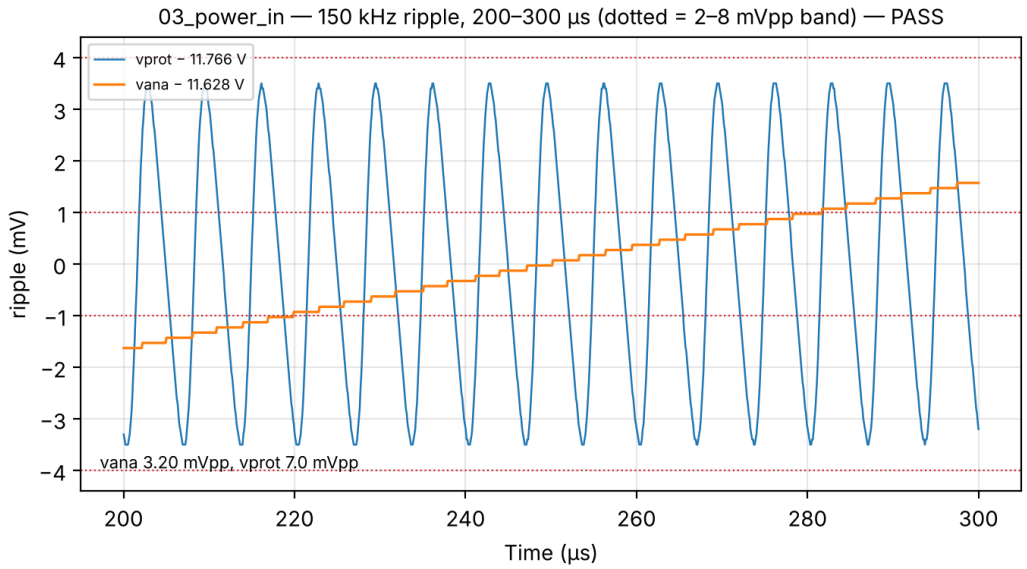


Figure 22: Ripple on the analog rail

Card 4: reverse supply

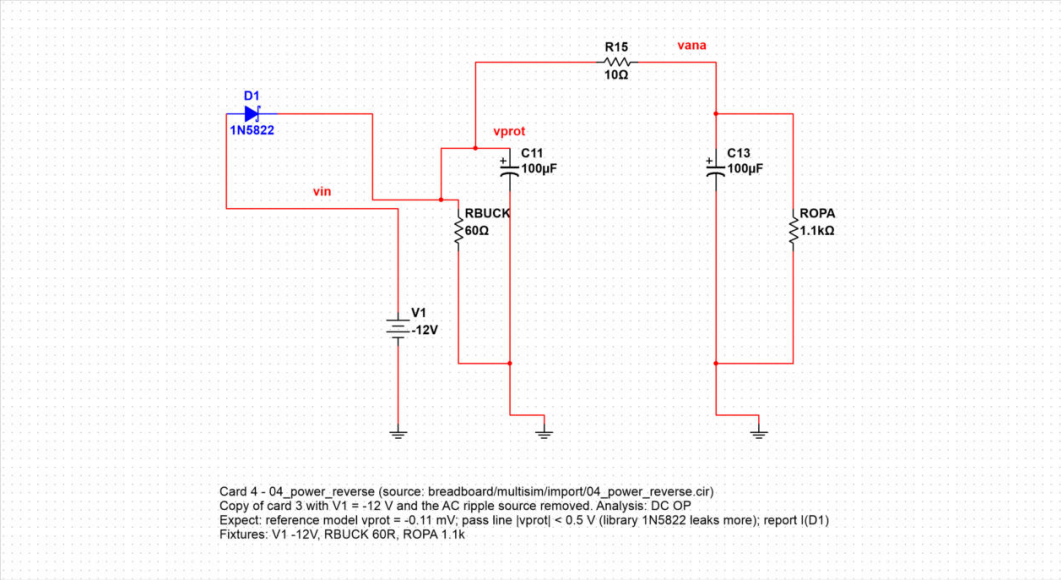


Figure 23: Multisim schematic

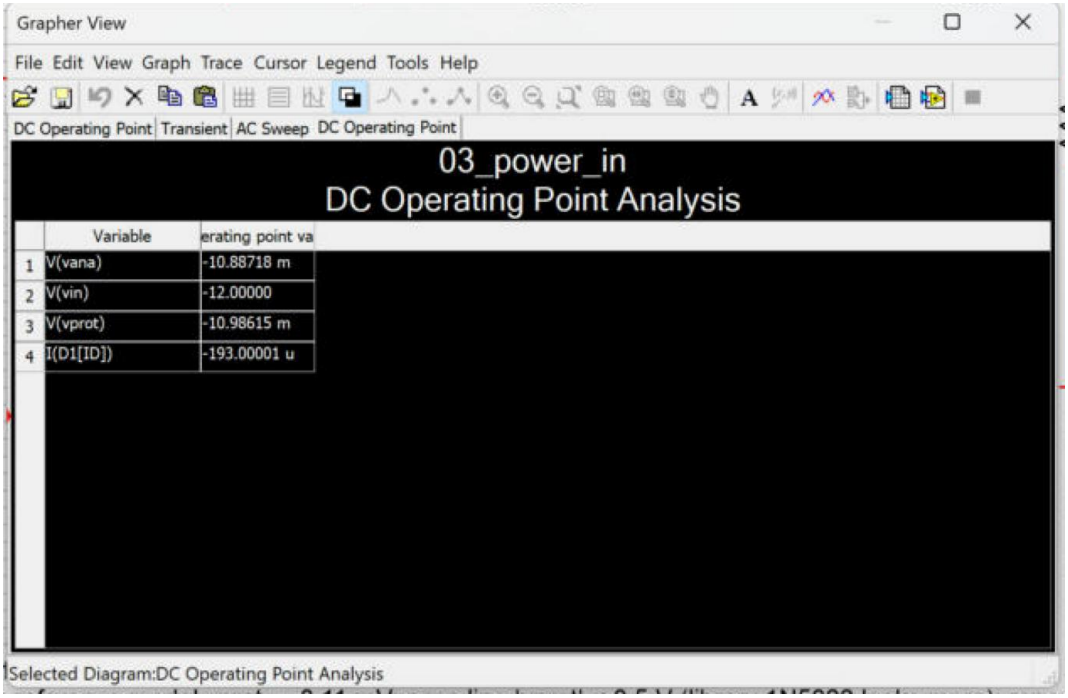


Figure 24: Multisim grapher, DC operating point

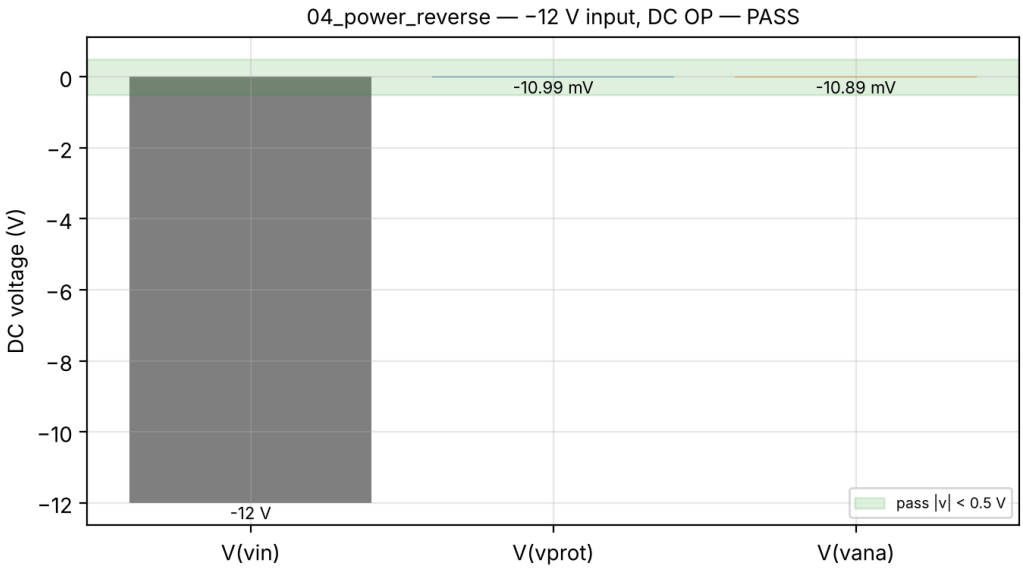


Figure 25: Reverse supply DC

Card 5: sync divider

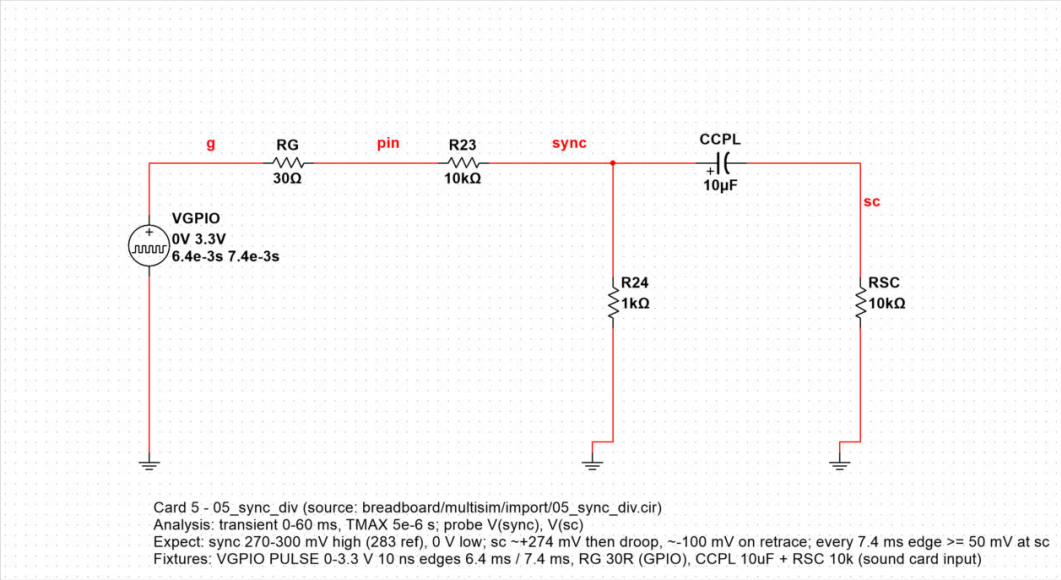


Figure 26: Multisim schematic

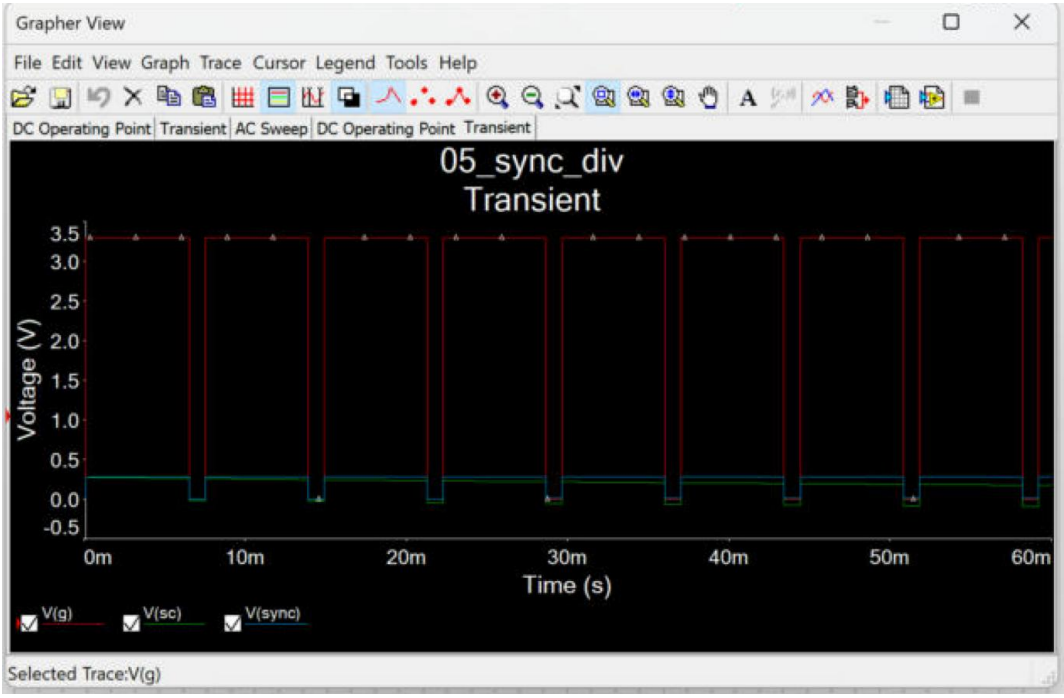


Figure 27: Multisim grapher, transient

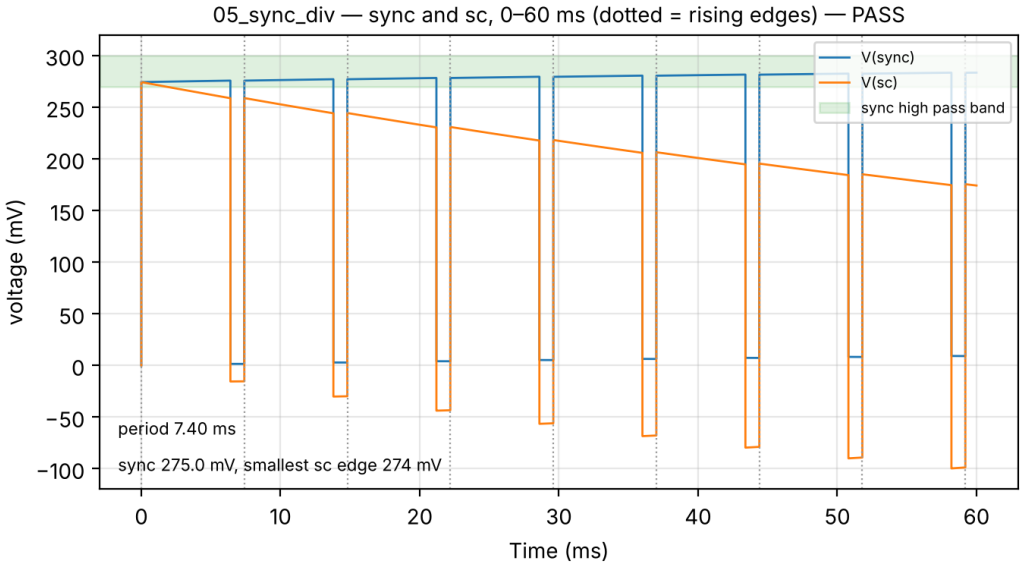


Figure 28: Sync pulse into the sound card

Card 6: IF input

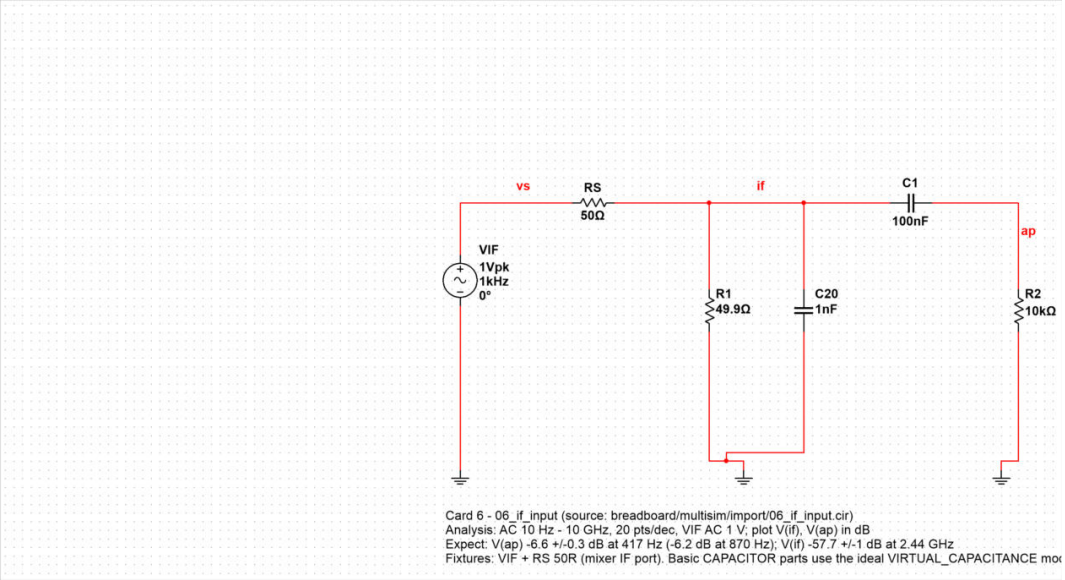


Figure 29: Multisim schematic

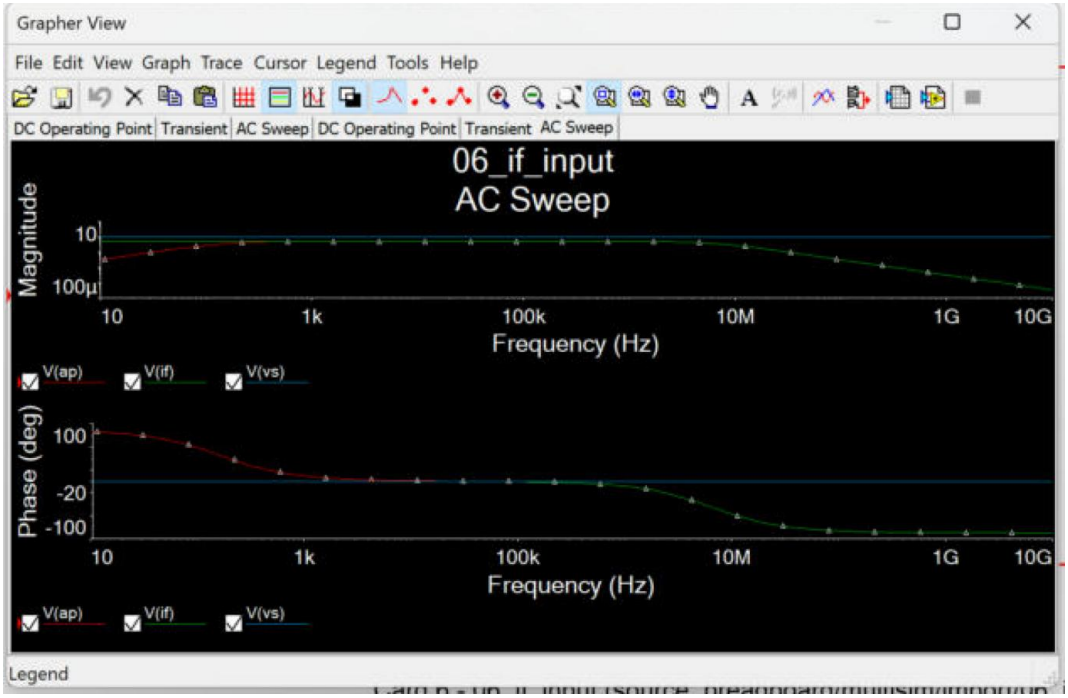


Figure 30: Multisim grapher, AC sweep

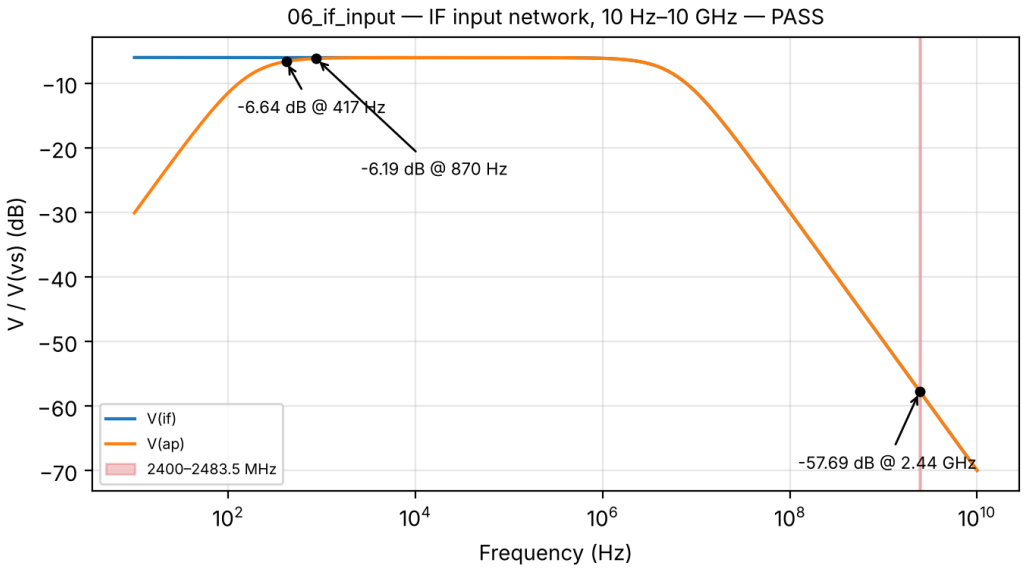


Figure 31: IF input response

Card 7: scaled radar

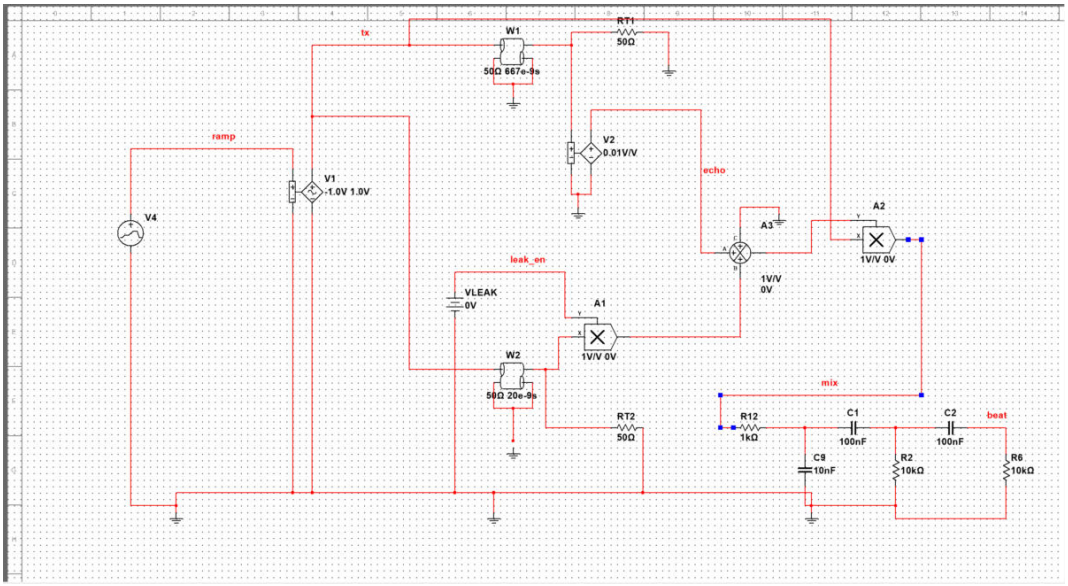


Figure 32: Multisim schematic

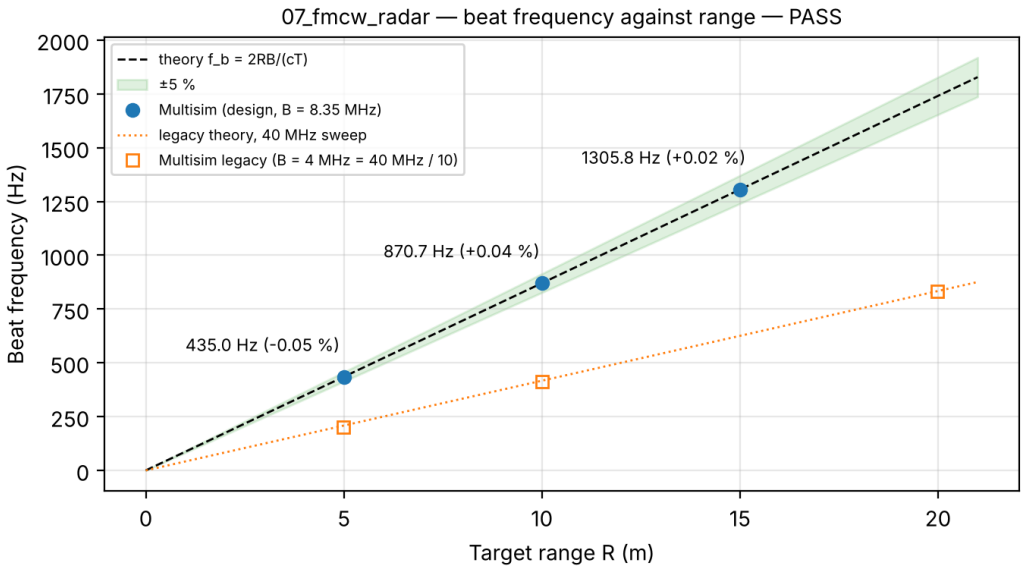


Figure 33: Beat frequency vs. range

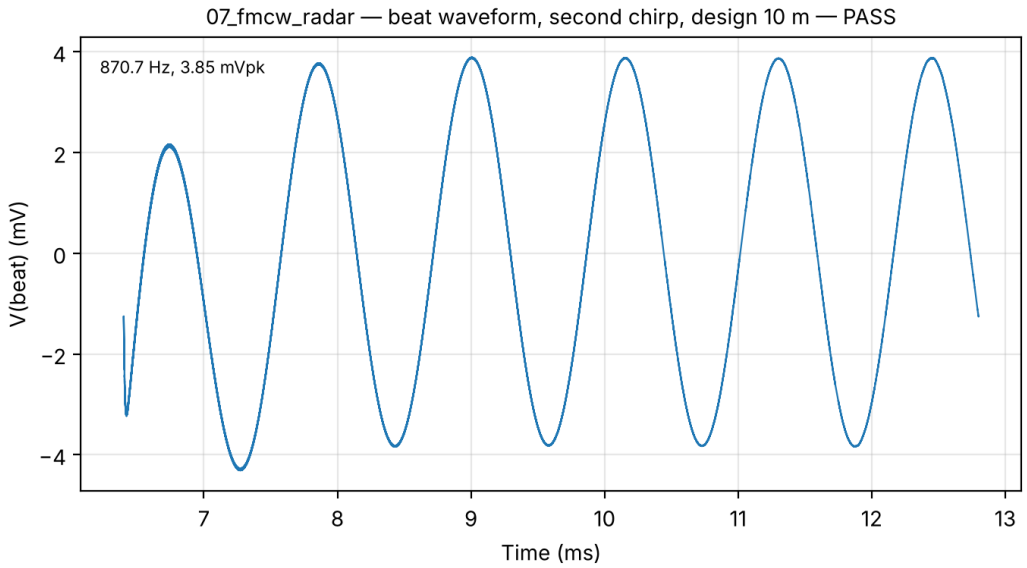


Figure 34: Beat waveform

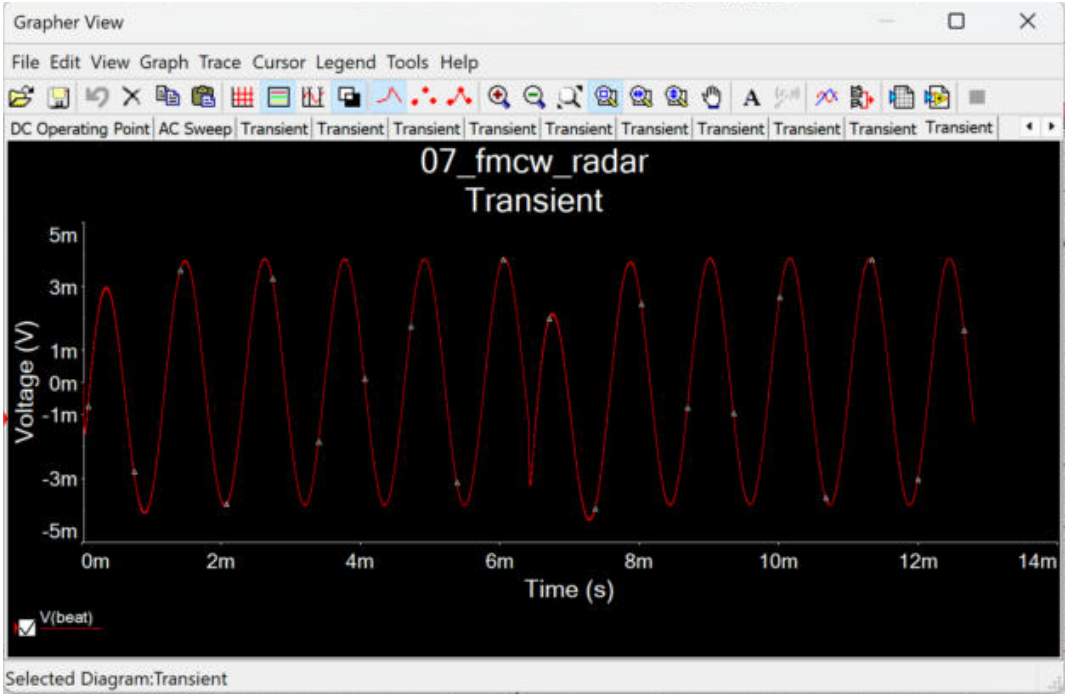


Figure 35: Multisim grapher, transient (10 m)

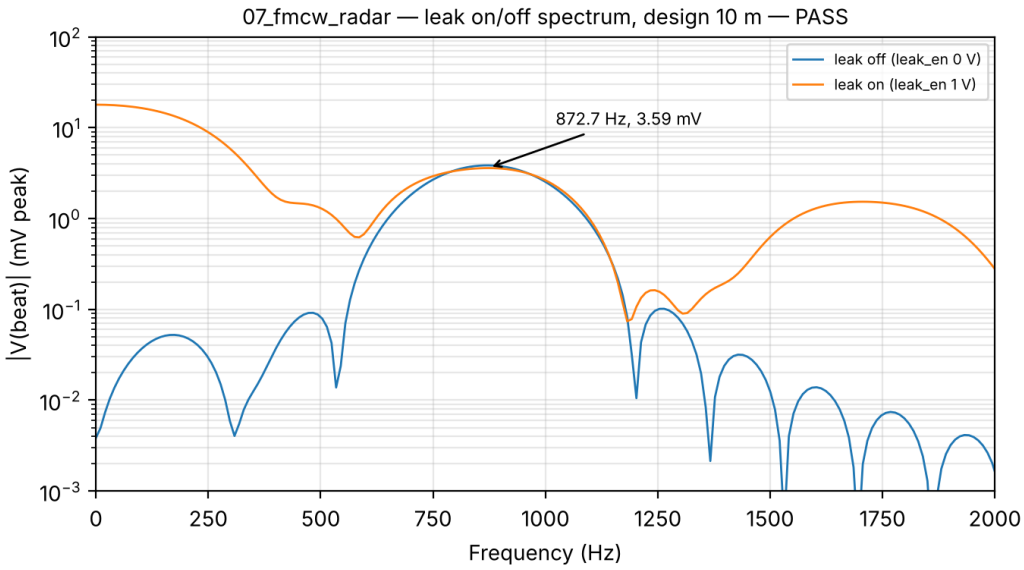


Figure 36: Spectrum with leakage on/off

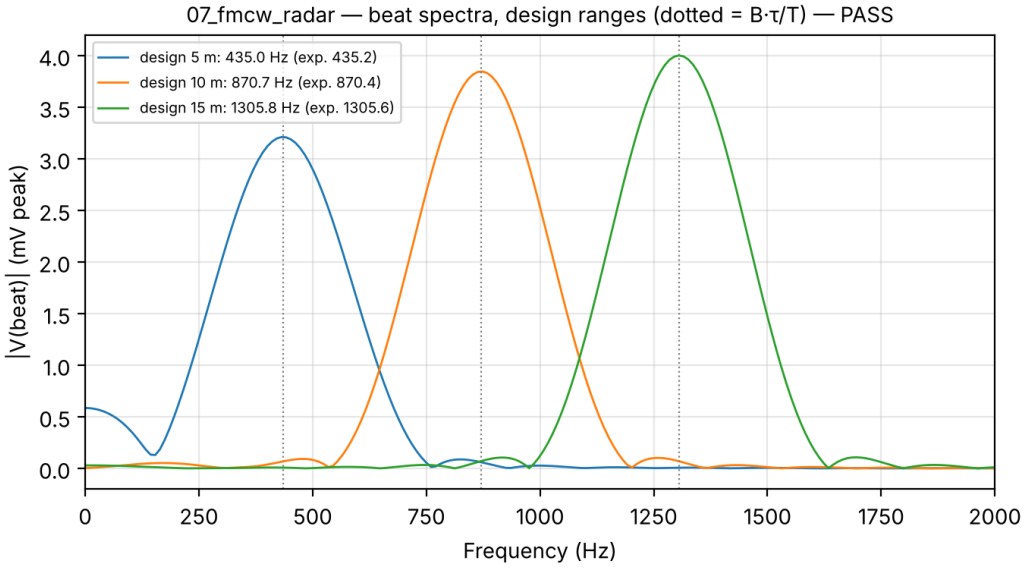


Figure 37: Spectra at each range

Card 8: all blocks together

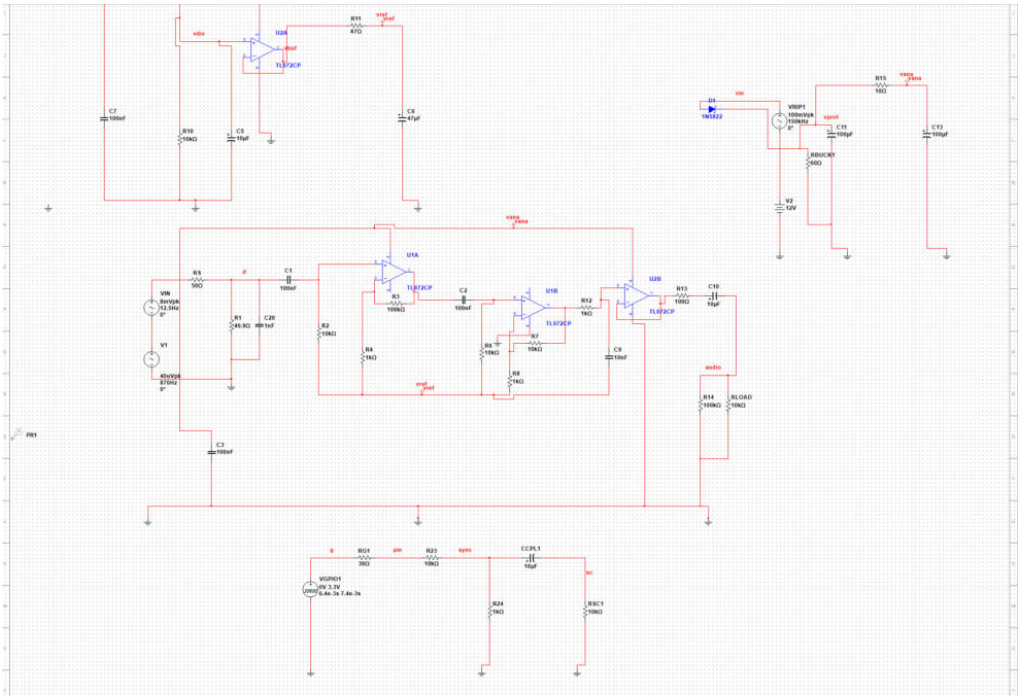


Figure 38: Multisim schematic (screenshot from before the supply-pin fix)

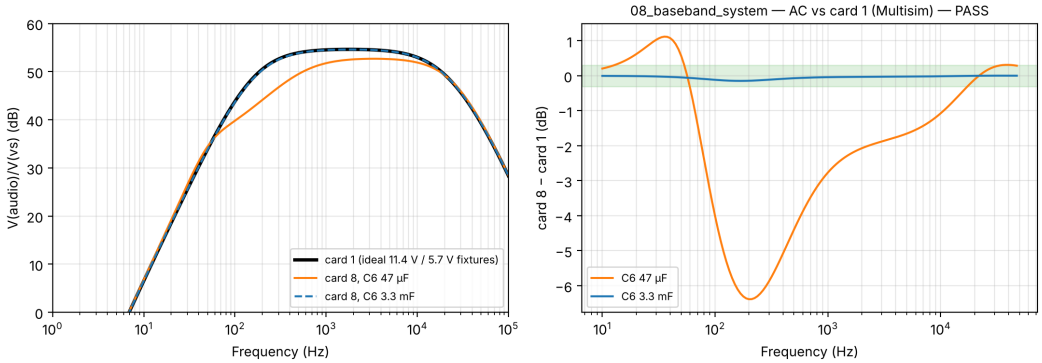


Figure 39: Gain vs. card 1, C6 47 µF and 3.3 mF

08_baseband_system — DC operating point, fixed sheet — PASS

node	DC operating point
V(am)	5.74711 V
V(aout)	5.74818 V
V(ap)	5.74710 V
V(audio)	0.00000 V
V(bout)	5.74722 V
V(bp)	5.74710 V
V(if)	0.00000 V
V(lp)	5.74722 V
V(obuf)	5.74723 V
V(pin)	0.00000 V
V(sc)	0.00000 V
V(sync)	0.00000 V
V(vana)	11.49418 V
V(vbuf)	5.74710 V
V(vdiv)	5.74709 V
V(vin)	12.00000 V
V(vprot)	11.72227 V
V(vref)	5.74710 V

Figure 40: DC operating point, all blocks (fixed)

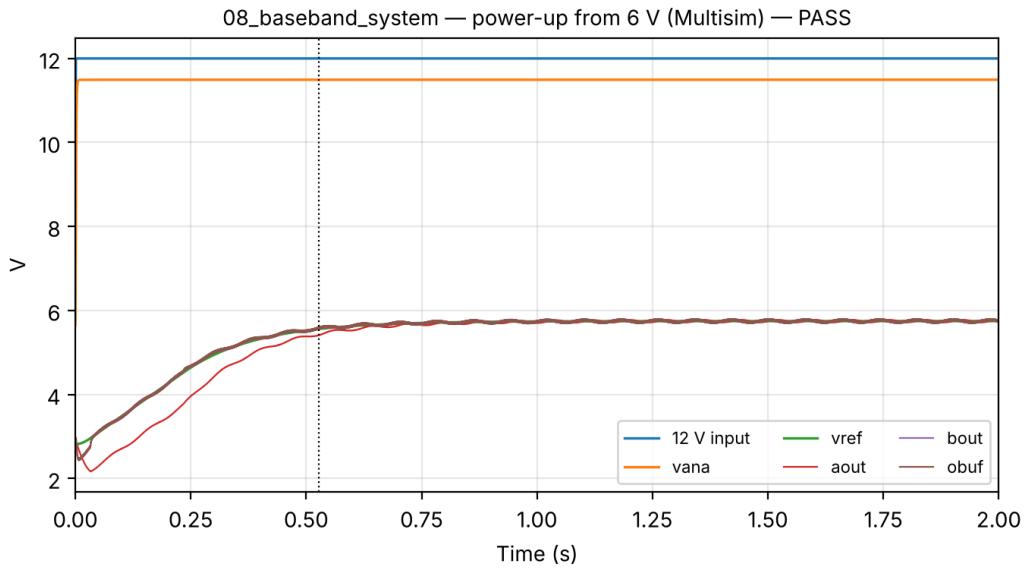


Figure 41: Power-up from 6 V

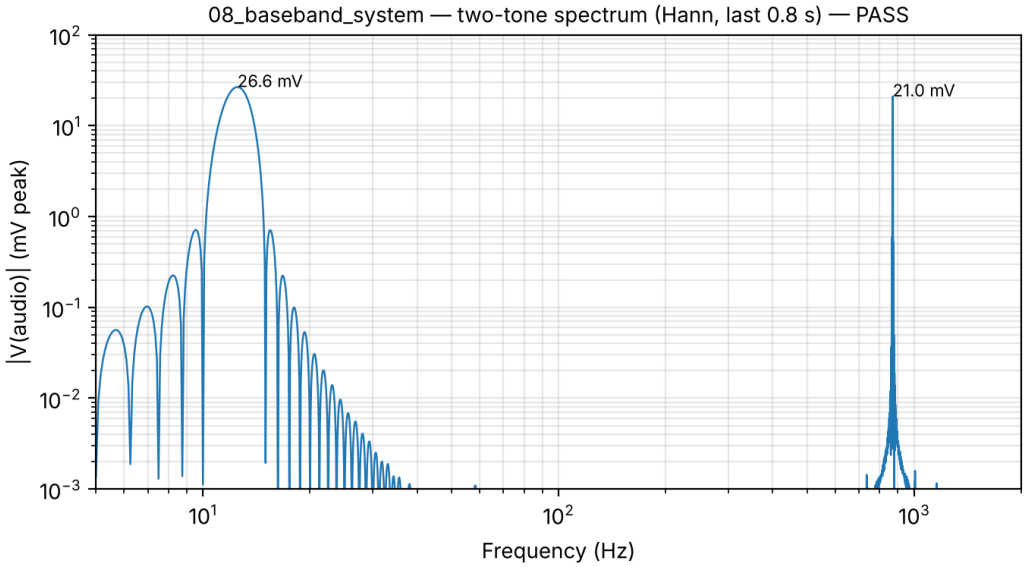


Figure 42: Two-tone spectrum, all blocks